

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 3: Neural Networks: Feedforward Networks and the Training Process

Student Name: _Monia

Nijimbere_____

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GitHub Repository Link: __<https://github.com/Nijimbere722/lab-3.git>

Date Submitted: __Le 28/07/2026_____

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	16%
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Claude	To understand the Concept I don't understand and guide to some errors that I am not sure	Can you explain how the Mean Squared Error loss is calculated and show how the gradients with respect to the weight and bias are derived? Please explain each step in simple language language	Implementing the forward and backward pass of a neural network using NumPy It also helped me understand the purpose of key components in the PyTorch training loop (such as <code>loss.backward()</code> , <code>optimizer.zero_grad()</code> , and <code>optimizer.step()</code>)	I used the AI responses as explanations rather than final answers. After receiving explanations, I compared them with the lab instructions and course materials to ensure they matched the requirements consistent with the expected behavior.


```

/usr/local/lib/python3.12/dist-
packages/numpy/_core/_methods.py:127: RuntimeWarning: overflow
encountered in reduce
    ret = umr_sum(arr, axis, dtype, out, keepdims, where=where)
/tmp/ipykernel_993/363528464.py:13: RuntimeWarning: overflow
encountered in square
    loss = np.mean((y_hat - y_col) ** 2)
/tmp/ipykernel_993/363528464.py:19: RuntimeWarning: overflow

```

Show more



for xb, yb in exp_train_loader:

 xb, yb = xb.to(DEVICE), yb.to(DEVICE)

 # ✗ you accidentally forgot this line

 # optimizer.zero_grad()

 logits = exp_model(xb)

 loss = criterion(logits, yb)

 loss.backward()

 optimizer.step()

Example 3 – Forgot `model.eval()`

</> Python

```
# ✗ Validation starts immediately  
with torch.no_grad():
```

Example 4 – Wrong output classes

</> Python

```
# ✗ Wrong output size  
layers.append(nn.Linear(prev_size, 6))
```

instead of

</> Python

```
layers.append(nn.Linear(prev_size, 7))
```



3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	yes	N/A
Prompt history	yes	N/A
Screenshot of AI interaction	yes	N/A
Exported AI conversation	yes	N/A
AI-generated code snippet	yes	N/A

Revised version showing student input	yes	N/A
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4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature: _____ Monia Nijimbere _____

Date: _____ 28/07/2026 _____